

Data Preprocessing

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Key Steps in Data Preprocessing

Data Cleaning



Loading and
inspecting data

STEP
01



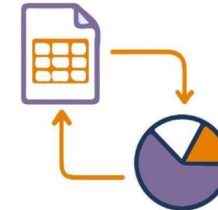
Deleting columns

STEP
02



Handling duplicates

STEP
03



Data type conversion

STEP
04



Handling missing data

STEP
05

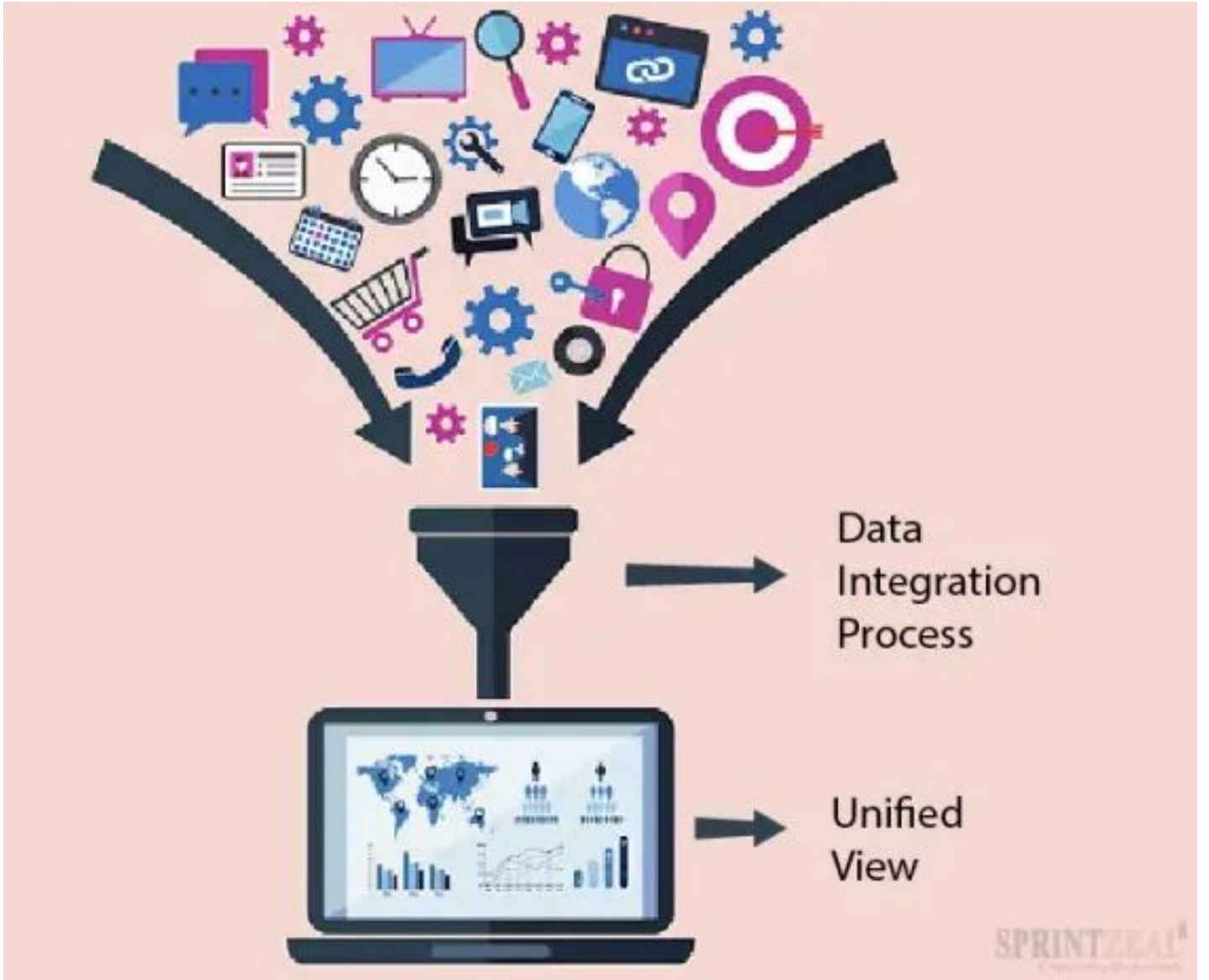


Dealing with outliers

STEP
06

DATA CLEANING STEPS

Data Integration



Data Transformation

- Standardization

The diagram illustrates the formula for standardization, $x' = \frac{x - \mu}{\sigma}$. It includes four labels with arrows pointing to the corresponding parts of the formula: 'Standardised Value' points to x' , 'Original Value' points to x , 'Sample Mean' points to μ , and 'Sample Standard Deviation' points to σ .

$$\text{Standardised Value } x' = \frac{\text{Original Value } x - \text{Sample Mean } \mu}{\text{Sample Standard Deviation } \sigma}$$

Data Transformati on

-
- Normalization
 - MinMaxScaling

$$x' = \frac{x - \min(x)}{\max(x) - \min(x)}$$

Encoding

- Categorical data
- Nominal ☐ No Relationship
- Ordinal ☐ Relationship

Index	Educational Qualifications
0	Masters
1	PhD
2	Diploma
3	High school
4	Bachelors
5	Masters
6	Bachelors



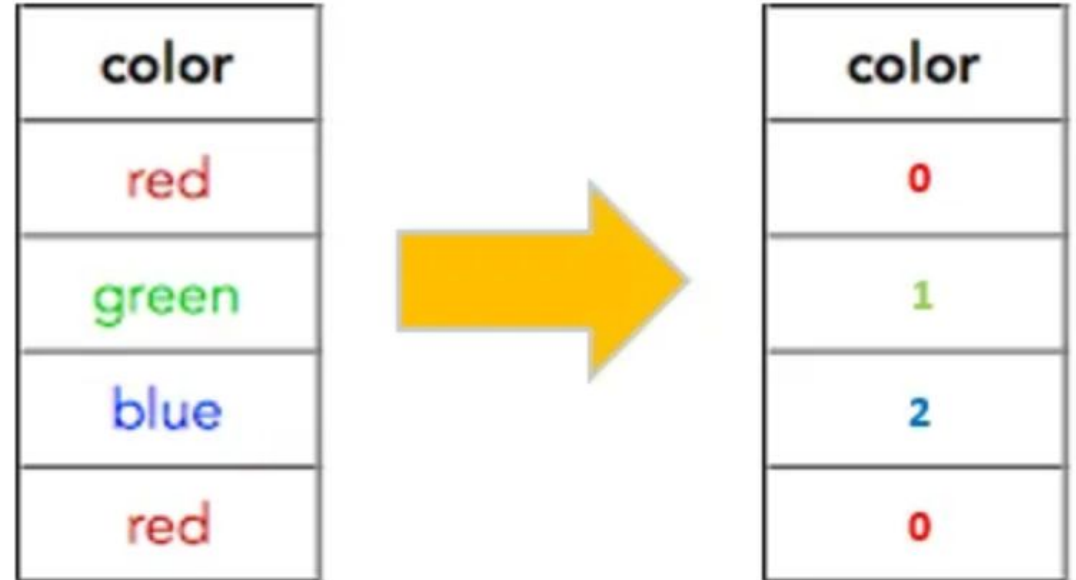
Index	Educational Qualifications
0	4
1	5
2	2
3	1
4	3
5	4
6	3

Ordinal Encoding

Work on input feature

Label Encoder

- Apply on Output Feature



One Hot Encoding

- for Nominal data

Original

Gender
Male
Female
Male
Male



One-hot encoded

Gender	Male	Female
Male	1	0
Female	0	1
Male	1	0
Male	1	0

Dummy Variable Trap

- We will remove our first columns (not necessary but recommended)
- Multicollinearity (should not math relationship in input)
- N-1

The diagram illustrates the process of encoding a categorical variable 'X' with 8 observations. The original data is shown in a table on the left. Two arrows point from this table to two separate encoding tables on the right. The top arrow, labeled 'One-Hot Encoding', points to a table with three columns: 'X = a', 'X = b', and 'X = c'. The bottom arrow, labeled 'Dummy Encoding', points to a table with two columns: 'X = a' and 'X = b'. In both encoding tables, the first column is 'id'.

id	X
1	a
2	c
3	a
4	b
5	a
6	c
7	c
8	b

id	X = a	X = b	X = c
1	1	0	0
2	0	0	1
3	1	0	0
4	0	1	0
5	1	0	0
6	0	0	1
7	0	0	1
8	0	1	0

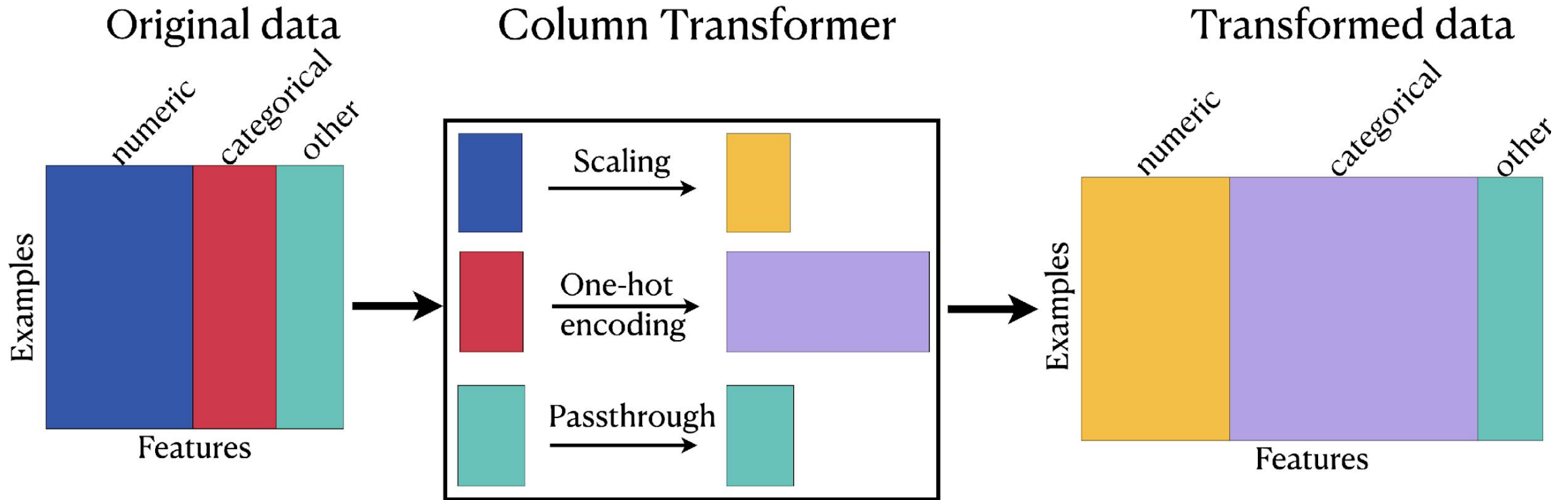
id	X = a	X = b
1	1	0
2	0	0
3	1	0
4	0	1
5	1	0
6	0	0
7	0	0
8	0	1

OHE using most frequent variables

- Pick top frequent categories.
- Replace others with "Other" or ignore.
- One-Hot Encode only these categories.

Column Transformer

- Avoids manual concatenation of processed features.
- Allows you to handle **mixed data types** easily

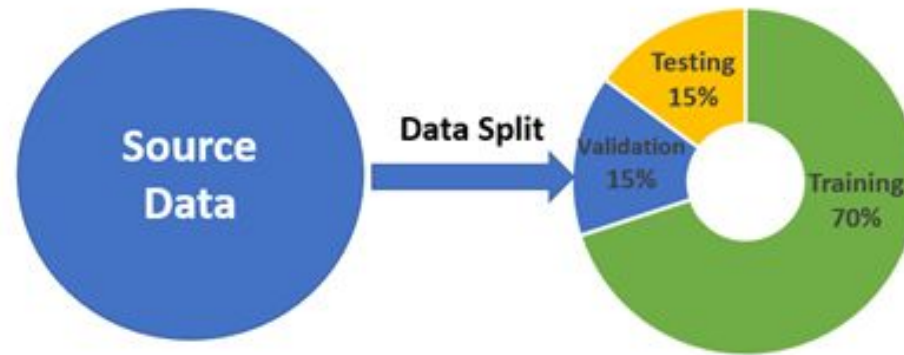


Data Reduction

- Reduce data size while keeping important info.

Techniques:

- Dimensionality reduction (PCA).
- Feature selection.
- Sampling.



Data Splitting

Split data into **training**, **validation**, and **test** sets.